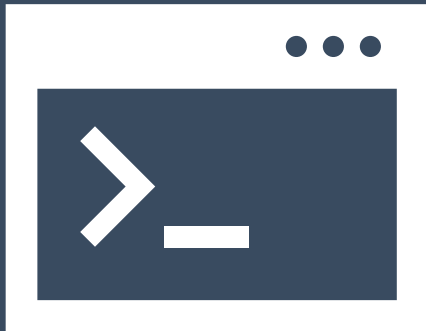


Introduction to Single-cell RNA-seq Analysis

<https://tinyurl.com/hbc-intro-scrna>



Harvard Chan Bioinformatics Core

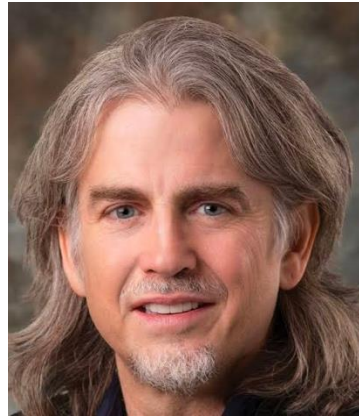


Introductions!





Shannan Ho Sui
Director



John Quackenbush
Faculty Advisor



James Billingsley



Emma Berdan



Elizabeth
Partan



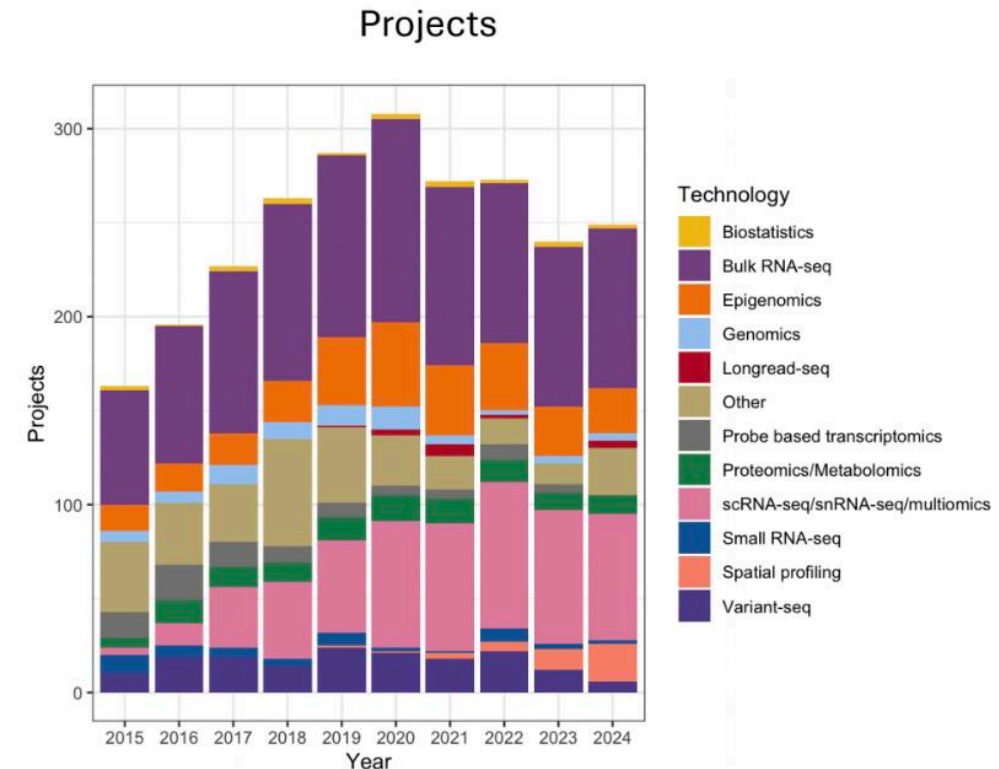
Will Gammerdinger



Noor Sohail

Consulting

- ❖ Transcriptomics: Bulk, single cell, small RNA
- ❖ Epigenomics: ChIP-seq, CUT&RUN, ATAC-seq, DNA methylation
- ❖ Variant discovery: WGS, resequencing, exome-seq and CNV
- ❖ Multiomics integration
- ❖ Spatial biology
- ❖ Experimental design and grant support



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HARVARD
T.H. CHAN
SCHOOL OF PUBLIC HEALTH

NIEHS



THE HARVARD CLINICAL
AND TRANSLATIONAL
SCIENCE CENTER



HARVARD
MEDICAL SCHOOL

Training

- ❖ Hands-on workshops design to reflect best practices, reproducibility and an emphasis on experimental design

- ❖ Basic Data Skills

- ❖ Shell

- ❖ R

- ❖ Python

- ❖ Advanced Topics: Analysis of high-throughput sequencing data

- ❖ Chromatin Biology

- ❖ Bulk RNA-seq

- ❖ Differential Gene Expression

- ❖ scRNA-seq

- ❖ Variant Calling

- ❖ Current Topics in Bioinformatics

<https://hsph.harvard.edu/research/bioinformatics/training/>

Training

❖ Hands-on workshops design to reflect best practices, reproducibility and an emphasis on experimental design

❖ Basic Data Skills

- ❖ Shell
- ❖ R
- ❖ Python

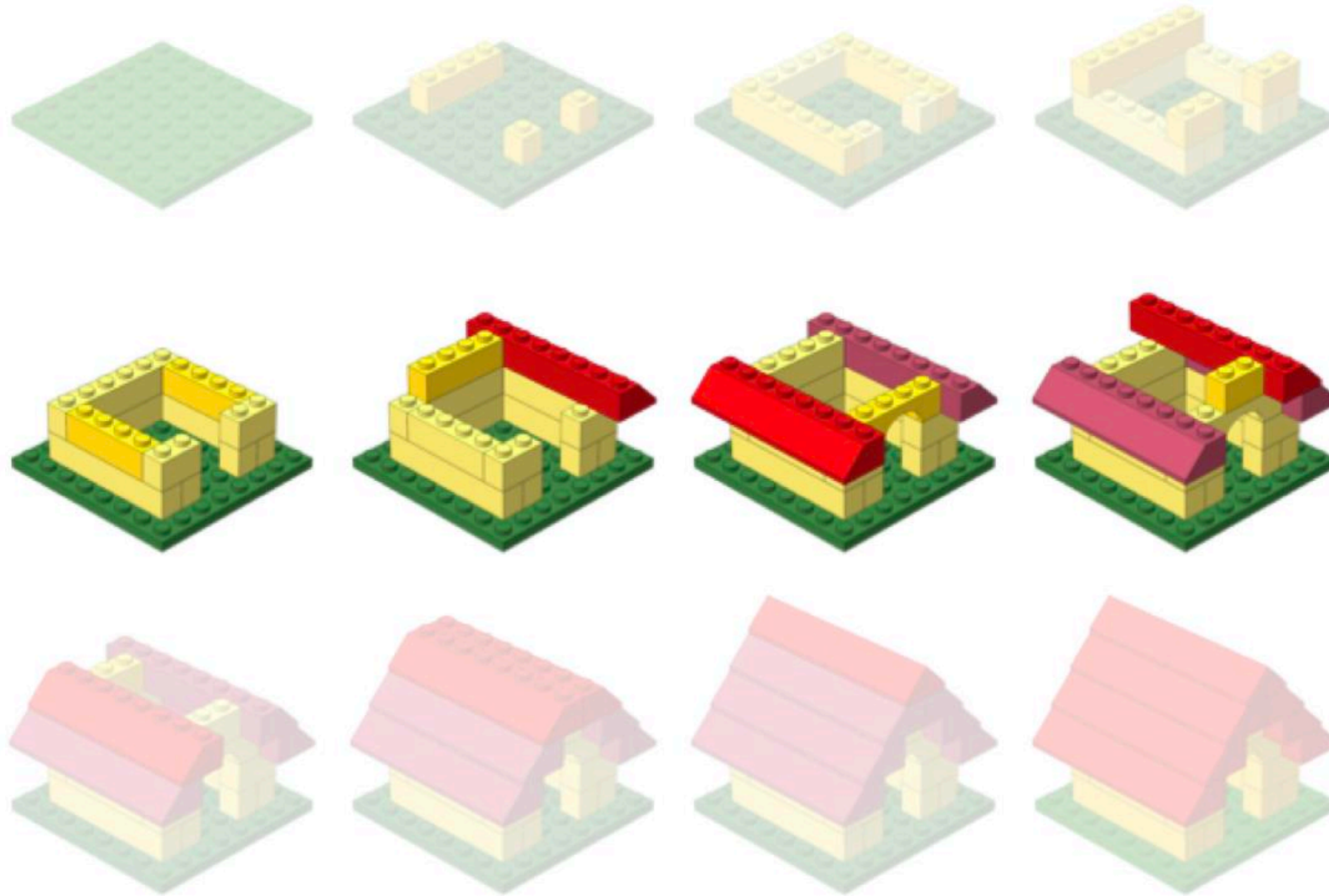
❖ Advanced Topics: Analysis of high-throughput sequencing data

- ❖ Chromatin Biology
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- ❖ Differential Gene Expression
- ❖ scRNA-seq
- ❖ Variant Calling

❖ Current Topics in Bioinformatics

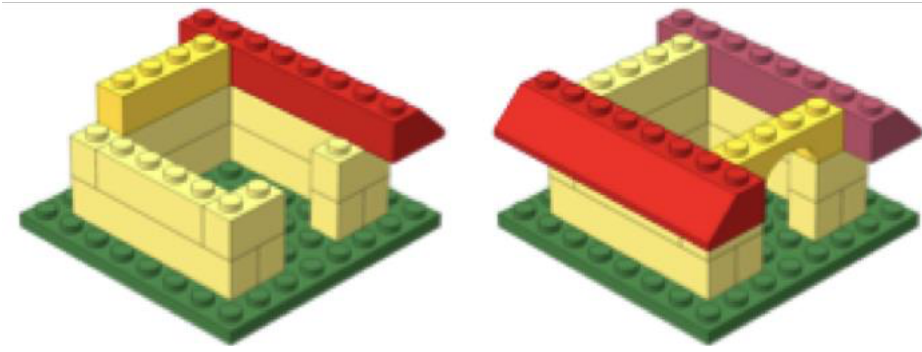


Workshop scope



Bioinformatic Data Analysis

Workshop Scope



- ❖ Describe best practices for designing a single-cell RNA-seq experiment
- ❖ Describe steps in a single-cell RNA-seq analysis workflow
- ❖ Use Seurat/Scanpy and associated tools to perform analysis of single-cell expression data, including data filtering, QC, integration, clustering, and marker identification
- ❖ Understand practical considerations for performing scRNA-seq, rather than in-depth exploration of algorithm theory

Logistics



Course schedule

Day 1

Time	Topic	Instructor
09:30 - 09:45	<u>Workshop introduction</u>	Will
09:45 - 10:35	<u>Bioinformatics begins at the bench: How experimental decisions shape single-cell RNA-seq analyses</u>	<u>Dr. Arpita Kulkarni</u>
10:35 - 10:40	Break	
10:40 - 11:00	scRNA-seq pre-reading discussion	Noor
11:00 - 11:45	<u>Quality control set-up</u>	Will
11:45 - 12:00	Overview of self-learning materials and homework submission	Will

<https://tinyurl.com/hbc-intro-scrna>

Course materials

❖ We continuously update our materials to reflect changes in the field/software

The screenshot displays the course website for 'Introduction to single-cell RNA-seq'. The top navigation bar includes links for 'Introduction to single-cell RNA-seq', 'Schedule', 'HBC', 'GitHub', 'Contact us', and a search icon. A left sidebar lists the course structure: 'Pre-reading:', 'Day 1:', 'Loading Single-cell RNA-seq Data' (highlighted), 'Day 1 Self-learning:', 'Day 2:', 'Day 2 Self-learning:', 'Day 3:', 'Day 3 Self-learning:', and 'Day 4:'. The main content area is titled 'Loading Single-cell RNA-seq Data' and includes a 'Code' button. Below the title are tags for 'SINGLE-CELL RNA-SEQ', 'QUALITY CONTROL', 'R', and 'PYTHON'. The text describes the lesson's focus on loading 10X Genomics data and generating objects for analysis. It lists keywords: 'data import, 10X Genomics, Seurat, scanpy' and states an approximate time of 90 minutes. A 'Learning objectives' section follows, listing three goals: setting up the environment, establishing data management practices, and demonstrating data import and quality control. A right sidebar titled 'On this page' provides a table of contents for the lesson, including 'Learning objectives', 'Overview of lesson', 'Exploring the example dataset', 'Set up', 'Data files used as input', 'Reading in a single sample', and 'Reading in multiple samples with a for loop'.

Introduction to single-cell RNA-seq Schedule HBC GitHub Contact us

Pre-reading: >
Day 1: >
Loading Single-cell RNA-seq Data
Day 1 Self-learning: >
Day 2: >
Day 2 Self-learning: >
Day 3: >
Day 3 Self-learning: >
Day 4: >

Loading Single-cell RNA-seq Data

</> Code

SINGLE-CELL RNA-SEQ QUALITY CONTROL R PYTHON

In this lesson, participants learn how to load single-cell RNA-seq count data. The lesson introduces the structure of typical 10X Genomics output files, demonstrates how to import these files efficiently and shows how to generate objects for one or multiple samples. Participants also practice using loops to streamline data loading and explore the metadata. By the end, participants will be able to prepare their data for downstream quality control and analysis.

KEYWORDS
data import, 10X Genomics, Seurat, scanpy

Approximate time: 90 minutes

Learning objectives

In this lesson, we will:

- Setup the environment for a single-cell analysis.
- Establish good data managements and metadata practices.
- Demonstrate how to import data and set up a project for upcoming quality control analysis.

On this page

- Learning objectives
- Overview of lesson
- Exploring the example dataset
- Set up
- Data files used as input
- Reading in a single sample
- Reading in multiple samples with a for loop

<https://tinyurl.com/hbc-intro-scrna>

Single Screen & 3 Windows

The image is a composite of three overlapping windows, illustrating a workflow for data analysis and learning.

Zoom Meeting Window: Shows a meeting with participants Noor Sohal and Will Gammerding. The chat window is open, displaying a message from NS: "I have a question about how to make figures in Python. How do you make multiple plots at once?".

JupyterLab Notebook Window: Displays a notebook titled "intro_python.py". The code includes:

```
metadata = pd.read_csv("data/mouse_exp_design.csv")

# Get the row names
metadata.index

# Get the column names
metadata.columns

Index(['sample1', 'sample2', 'sample3', 'sample4', 'sample5', 'sample6',
       'sample7', 'sample8', 'sample9', 'sample10', 'sample11', 'sample12'],
      dtype='object')

metadata.loc[0, :]

genotype    WT
celltype     typeA
replicate     3
Name: sample1, dtype: object

metadata.loc[:, 0]

sample1    WT
sample2    WT
sample3    WT
sample4    KO
sample5    KO
sample6    KO
sample7    WT
sample8    WT
sample9    WT
sample10   KO
sample11   KO
sample12   KO
Name: genotype, dtype: object
```

Web Browser Window: Shows a page titled "Python and". The sidebar lists learning objectives: Overview of lesson, What is Python?, Anaconda, JupyterLab, and Creating an interactive notebook. The main content area displays a box plot titled "differences in average gene expression".

Box Plot: The y-axis is labeled "Mean expression" and ranges from 10 to 14. The x-axis is labeled "Genotype" with categories "KO" and "WT". The legend indicates "celltype" with "typeA" in blue and "typeB" in orange. The plot shows that for the KO genotype, typeB has a higher mean expression (around 13.5) compared to typeA (around 10.5). For the WT genotype, typeA has a higher mean expression (around 10.5) compared to typeB (around 10.2).

Our Recommendation

Single Screen & 3 Windows

The image is a composite of three overlapping windows on a single screen, demonstrating a workflow for data analysis and learning.

Zoom Meeting Window: Shows a video call with participants Noor Sohal and Will Gammerdinger. The Zoom logo is prominently displayed.

JupyterLab Notebook Window: Displays a Python notebook with the following code and output:

```
[28]: metadata = pd.read_csv("data/mouse_exp_design.csv")

[29]: # Get the row names
metadata.index
# Get the column names
metadata.columns

[30]: Index(['sample1', 'sample2', 'sample3', 'sample4', 'sample5', 'sample6',
'sample7', 'sample8', 'sample9', 'sample10', 'sample11', 'sample12'],
dtype='object')

[31]: metadata.loc[0, :]

[32]: genotype    WT
celltype        typeA
replicate        3
Name: sample1, dtype: object

[33]: metadata.loc[:, 0]

[34]: sample1    WT
sample2    WT
sample3    WT
sample4    KO
sample5    KO
sample6    KO
sample7    WT
sample8    WT
sample9    WT
sample10   KO
sample11   KO
sample12   KO
Name: genotype, dtype: object
```

Web Browser Window: Shows a page titled "Python and" with a sidebar containing "Learning objectives", "Overview of lesson", "What is Python?", "Anaconda", "JupyterLab", and "Creating an interactive Notebook". The main content area displays a box plot titled "Differences in average gene expression" comparing mean expression across genotypes (KO and WT) for cell types (typeA and typeB).

Box Plot Data (Estimated):

Genotype	Celltype	Mean Expression (approx.)
KO	typeA	10.5
	typeB	10.8
WT	typeA	10.2
	typeB	12.5

Our Recommendation

Single Screen & 3 Windows

The image illustrates a single-screen setup for a Zoom meeting, showing three overlapping windows. The top-left window displays a JupyterLab interface with a file explorer on the left and a code editor on the right. The top-right window shows a web browser with a page titled "Python and JupyterLab for beginners". The bottom window shows a box plot titled "Differences in average gene expression" comparing mean expression across genotypes (KO and WT) for two cell types (typeA and typeB). A green border highlights the three windows.

Web Browser

Differences in average gene expression

Mean expression

Genotype

celltype

typeA

typeB

KO

WT

Our Recommendation

Single Screen & 3 Windows

Our Recommendation

Jupyter Notebook

Mean expression

Genotype

celltype

typeA

typeB

KO

WT

Single Screen & 3 Windows

Zoom

Web Browser

Jupyter Notebook

Our Recommendation

The image displays three overlapping windows on a dark blue background. The top-left window is a Zoom meeting interface showing a video call with participants Noor Sohal and Will Gammerdinger. The top-right window is a web browser displaying a page titled 'Python and JupyterLab for beginners'. The bottom window is a Jupyter Notebook interface showing a Python script for reading and analyzing a CSV file. A large green text 'Web Browser' is overlaid on the browser window, and a large purple text 'Jupyter Notebook' is overlaid on the Jupyter Notebook window. A large yellow text 'Our Recommendation' is overlaid on the bottom left of the image.

Course participation

- ❖ Mandatory review of self-learning lessons and assignments
- ❖ Attendance required for all classes
- ❖ Your questions and active participation drive learning
- ❖ **We look forward to all of your questions!**



Course participation

- ❖ At-home lessons and exercises after each session
- ❖ Cover material not previously discussed
- ❖ Provides us feedback to help pace the course appropriately
- ❖ 3-5 hours to complete

Using AI for Assignments

❖ Do

- ❖ Try to resolve error messages with it
- ❖ Test code written by AI on a dataset where you have expected results
- ❖ Take the time to review the generated code line-by-line

❖ Don't

- ❖ Implement it in replacement to learning
- ❖ Write code that you don't understand
- ❖ Assume the output from an AI process is correct

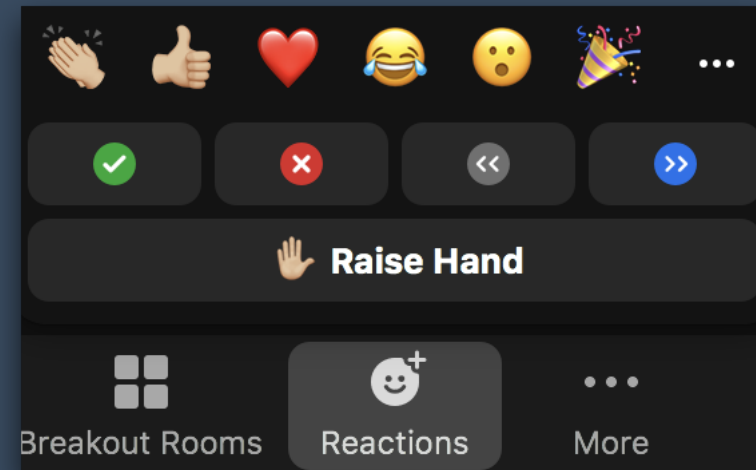
Odds & Ends

❖ Quit/minimize all applications that are not required for class

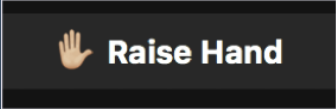
❖ Are you all set?

❖  = "agree", "I'm all set"

❖  = "disagree", "I need help"



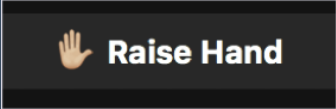
Odds & Ends

- ❖ Questions for the presenter?
 - ❖ Post the question in the Chat window OR
 - ❖  when the presenter asks for questions
 - ❖ Let the Moderator know

Odds & Ends

❖ Questions for the presenter?

- ❖ Post the question in the Chat window OR

- ❖  when the presenter asks for questions

- ❖ Let the Moderator know

❖ Technical difficulties with software?

- ❖ Start a private chat with the Moderator with a description of the problem

Contact Us

- ❖ *HBC training team:* hbctraining@hsph.harvard.edu
- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu